Q1. JEE Main 2026 (21 January Shift 2)

For the matrices $\mathrm{A}=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and $\mathrm{B}=\begin{bmatrix}-29 & 49\\ -13 & 18\end{bmatrix}$, if $\left(\mathrm{A}^{15}+\mathrm{B}\right)\begin{bmatrix}x\\ y\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$, then among the following which one is true?

(1) $x=5, y=7$
(2) $x=18, y=11$
(3) $x=11, y=2$
(4) $x=16, y=3$

Q2. JEE Main 2026 (23 January Shift 1)

Let $|\mathrm{A}|=6$, where A is a 3×3 matrix. If $\left|\operatorname{adj}\left(3\operatorname{adj}\left(\mathrm{A}^2\cdot\operatorname{adj}(2\mathrm{A})\right)\right)\right|=2^{\mathrm{m}}\cdot 3^{\mathrm{n}}, \mathrm{m},\mathrm{n}\in\mathbf{N}$, then $\mathrm{m}+\mathrm{n}$ is equal to ____.

Q3. JEE Main 2026 (28 January Shift 2)

Let $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and B be two matrices such that $A^{100}=100B+I$. Then the sum of all the elements of B^{100} is ____

Q4. JEE Main 2026 (24 January Shift 2)

Let $f(x)=\int\frac{7x^{10}+9x^8}{(1+x^2+2x^9)^2}dx, x>0, \lim_{x\to 0}f(x)=0$ and $f(1)=\frac{1}{4}$. If $\mathrm{A}=\begin{bmatrix}0 & 0 & 1\\ \frac{1}{4} & f'(1) & 1\\ \alpha^2 & 4 & 1\end{bmatrix}$ and $\mathrm{B}=\operatorname{adj}(\operatorname{adj}\mathrm{A})$ be such that $|\mathrm{B}|=81$, then α^2 is equal to

(1) 2
(2) 1
(3) 3
(4) 4

Q5. JEE Main 2026 (24 January Shift 2)

Let $P=[p_{ij}]$ and $Q=[q_{ij}]$ be two square matrices of order 3 such that $q_{\mathrm{ij}}=2^{(\mathrm{i}+\mathrm{j}-1)}\mathrm{p}_{\mathrm{ij}}$ and $\det(\mathrm{Q})=2^{10}$. Then the value of $\det(\operatorname{adj}(\operatorname{adj}\mathrm{P}))$ is:

(1) 81
(2) 16
(3) 124
(4) 32

Q6. JEE Main 2026 (22 January Shift 2)

If $X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$ is a solution of the system of equations $AX=B$, where $\operatorname{adj}A=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & 5\\ 1 & -2 & 3\end{bmatrix}$ and $\mathrm{B}=\begin{bmatrix}4\\ 0\\ 2\end{bmatrix}$, then $|x+y+z|$ is equal to :

(1) 2
(2) $\frac{3}{2}$
(3) 1
(4) 3

Q7. JEE Main 2026 (22 January Shift 1)

If $\mathrm{A}=\begin{bmatrix}2 & 3\\ 3 & 5\end{bmatrix}$, then the determinant of the matrix $\left(\mathrm{A}^{2025}-3\,\mathrm{A}^{2024}+\mathrm{A}^{2023}\right)$ is

(1) 12 (2) 28 (3) 24 (4) 16

Q8. JEE Main 2026 (22 January Shift 1)

Let A be a 3×3 matrix such that $\mathrm{A}+\mathrm{A}^{\mathrm{T}}=\mathrm{O}$. If $\mathrm{A}\begin{bmatrix}1\\ -1\\ 0\end{bmatrix}=\begin{bmatrix}3\\ 3\\ 2\end{bmatrix}$, $\mathrm{A}^2\begin{bmatrix}1\\ -1\\ 0\end{bmatrix}=\begin{bmatrix}-3\\ 19\\ -24\end{bmatrix}$ and $\det(\operatorname{adj}(2\operatorname{adj}(\mathrm{A}+\mathrm{I})))=(2)^{\alpha}\cdot(3)^{\beta}\cdot(11)^{\gamma}$, α,β,γ are non-negative integers, then $\alpha+\beta+\gamma$ is equal to ____

Q9. JEE Main 2026 (23 January Shift 2)

Let $A=\begin{bmatrix}0 & 2 & -3\\ -2 & 0 & 1\\ 3 & -1 & 0\end{bmatrix}$ and B be a matrix such that $B(I-A)=I+A$. Then the sum of the diagonal elements of $\mathrm{B}^{\mathrm{T}}\mathrm{B}$ is equal to ____.

Q10. JEE Main 2026 (21 January Shift 1)

For some $\alpha,\beta\in\mathbf{R}$, let $A=\begin{bmatrix}\alpha & 2\\ 1 & 2\end{bmatrix}$ and $B=\begin{bmatrix}1 & 1\\ 1 & \beta\end{bmatrix}$ be such that $A^2-4A+2I=B^2-3B+I=O$. Then $\left(\det\left(\operatorname{adj}\left(A^3-B^3\right)\right)\right)^2$ is equal to ____.

Q11. JEE Main 2026 (28 January Shift 1)

Let A, B and C be three 2×2 matrices with real entries such that $B=(I+A)^{-1}$ and $\mathrm{A}+\mathrm{C}=\mathrm{I}$. If $\mathrm{BC}=\begin{bmatrix}1 & -5\\ -1 & 2\end{bmatrix}$ and $\mathrm{CB}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}12\\ -6\end{bmatrix}$, then x_1+x_2 is

(1) 2 (2) -2 (3) 0 (4) 4

Q12. JEE Main 2026 (24 January Shift 1)

The number of 3×2 matrices A, which can be formed using the elements of the set $\{-2,-1,0,1,2\}$ such that the sum of all the diagonal elements of $\mathrm{A}^{\mathrm{T}}\mathrm{A}$ is 5, is

ANSWERS AND SOLUTIONS

1. (3)	**2.** 62	**3.** 0	**4.** (4)	**5.** (2)	**6.** (1)	**7.** (4)	**8.** 18
9. 3	**10.** 225	**11.** (3)	**12.** 312				

1. **(3)** Here $A^n = \begin{bmatrix} 2n+1 & -4n \\ n & -2n+1 \end{bmatrix}$

$\Rightarrow A^{15} = \begin{bmatrix} 31 & -60 \\ 15 & -29 \end{bmatrix}$

$\Rightarrow A^{15} + B = \begin{bmatrix} 2 & -11 \\ 2 & -11 \end{bmatrix}$

Now $(A^{15} + B)\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 2 & -11 \\ 2 & -11 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow 2x - 11y = 0 \Rightarrow x = \frac{11y}{2}$.

For $y = 2$, $x = 11$.

2. **(62)** For 3×3 matrix: $|\text{adj}(M)| = |M|^2$, $|kM| = k^3|M|$.

$|2A| = 8 \cdot 6 = 48$, $|\text{adj}(2A)| = 48^2 = 2^8 \cdot 3^2$.

$|A^2 \cdot \text{adj}(2A)| = |A|^2 \cdot |\text{adj}(2A)| = 2^2 \cdot 3^2 \cdot 2^8 \cdot 3^2 = 2^{10} \cdot 3^4$.

Let $B = A^2 \cdot \text{adj}(2A)$. $|\text{adj}(B)| = |B|^2 = 2^{20} \cdot 3^8$.

$|3\,\text{adj}(B)| = 3^3 \cdot 2^{20} \cdot 3^8 = 2^{20} \cdot 3^{11}$.

$|\text{adj}(3\,\text{adj}(B))| = (2^{20} \cdot 3^{11})^2 = 2^{40} \cdot 3^{22}$.

$m = 40, n = 22 \Rightarrow m + n = 62$.

3. **(0)** $A = I + \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix}$, let $M = \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix}$

$M^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = M^3 = M^4 = \ldots = M^{100}$

$A^{100} = (I + M)^{100} = \sum_{r=0}^{100} \binom{100}{r} M^r \cdot I$

$A^{100} = I + 100M = I + 100B$

$\therefore M = B \Rightarrow M^{100} = B^{100} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

4. **(4)** Divide numerator and denominator by x^{18}, let $t = x^{-9} + x^{-7} + 2$, then $dt = -(9x^{-10} + 7x^{-8})dx$.

$f(x) = \frac{1}{t} + C = \frac{x^9}{1+x^2+2x^9} + C$.

$\lim_{x\to 0} f(x) = 0 \Rightarrow C = 0$. $f(1) = \frac{1}{4}$ (verified).

$f'(x) = \frac{9x^8(1+x^2+2x^9) - x^9(2x+18x^8)}{(1+x^2+2x^9)^2}$.

$f'(1) = \frac{36-20}{16} = 1$.

$$|A| = \begin{vmatrix} 0 & 0 & 1 \\ 1/4 & 1 & 1 \\ \alpha^2 & 4 & 1 \end{vmatrix} = 1 - \alpha^2.$$

$|B| = |\text{adj}(\text{adj } A)| = |A|^{(3-1)^2} = (1-\alpha^2)^4 = 81 = 3^4.$
$|1-\alpha^2| = 3 \Rightarrow \alpha^2 = 4.$

5. **(2)** $\begin{vmatrix} 2p_{11} & 2^2p_{12} & 2^3p_{13} \\ 2^2p_{21} & 2^3p_{22} & 2^4p_{23} \\ 2^3p_{31} & 2^4p_{32} & 2^5p_{33} \end{vmatrix} = 2^{10}$

$$2^2 \cdot 2 \cdot 2^3 \begin{vmatrix} p_{11} & p_{12} & p_{13} \\ 2p_{21} & 2p_{22} & 2p_{23} \\ 2^2p_{31} & 2^2p_{32} & 2^2p_{33} \end{vmatrix} = 2^{10}$$

$$2^9 \begin{vmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{vmatrix} = 2^{10} \Rightarrow |P| = 2$$

$|\text{adj}(\text{adj}(P))| = |P|^{(n-1)^2} = |P|^4 = 2^4 = 16$

6. **(1)** $X = \frac{1}{|A|}\text{adj}(A) \cdot B.$

$|\text{adj}(A)| = 4(10) - 2(-20) + 2(10) = 100$, so $|A|^2 = 100 \Rightarrow |A| = \pm 10.$

$$\text{adj}(A) \cdot B = \begin{pmatrix} 16+0+4 \\ -20+0+10 \\ 4+0+6 \end{pmatrix} = \begin{pmatrix} 20 \\ -10 \\ 10 \end{pmatrix}$$

$x + y + z = \frac{20-10+10}{|A|} = \frac{20}{\pm 10} = \pm 2$

$|x + y + z| = 2$

7. **(4)** $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 13 & 21 \\ 21 & 34 \end{bmatrix}$

$|A^{2025} - 3A^{2024} + A^{2023}|$
$= \left|A^{2023}\left(A^2 - 3A + I\right)\right|$
$= |A|^{2023}|A^2 - 3A + I|$
$= 1 \cdot \begin{vmatrix} 8 & 12 \\ 12 & 20 \end{vmatrix} = 160 - 144 = 16$

8.
(18) A is skew-symmetric: $A = \begin{bmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{bmatrix}.$

From $A\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$: $c = 3$, $a + b = -2.$

From $A^2\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 19 \\ -24 \end{bmatrix}$: $b = 3$, $a = -5.$

$A+I=\begin{bmatrix}1 & -3 & 3\\ 3 & 1 & 5\\ -3 & -5 & 1\end{bmatrix}$, $\det(A+I)=44$.

For 3×3 matrix: $\det(\operatorname{adj}(M))=(\det M)^2$.

$\det(2\operatorname{adj}(A+I))=8\cdot 44^2=8\cdot 1936=15488=2^7\cdot 11^2$.

$\det(\operatorname{adj}(2\operatorname{adj}(A+I)))=15488^2=2^{14}\cdot 11^4$.

$\alpha=14,\ \beta=0,\ \gamma=4$. $\alpha+\beta+\gamma=18$.

9. **(3)** A is skew-symmetric ($A^T=-A$).

From $B(I-A)=I+A$:

$B=(I+A)(I-A)^{-1}$.

$B^T=((I-A)^T)^{-1}(I+A)^T=(I+A)^{-1}(I-A)$.

Since A is skew-symmetric, $(I-A)$ and $(I+A)$ commute:

$(I-A)(I+A)=I-A^2=(I+A)(I-A)$.

$B^TB=(I+A)^{-1}(I-A)(I+A)(I-A)^{-1}=(I+A)^{-1}(I+A)(I-A)(I-A)^{-1}=I$.

Trace of $B^TB=\operatorname{tr}(I)=3$.

10. **(225)** From $A^2-4A+2I=O$: characteristic equation is $\lambda^2-4\lambda+2=0$.

For A: $\alpha+2=4\Rightarrow\alpha=2$.

From $B^2-3B+I=O$: characteristic equation is $\lambda^2-3\lambda+1=0$.

For B: $\beta+1=3\Rightarrow\beta=2$.

$A^3=14A-8I$, $B^3=8B-3I$.

$A^3-B^3=14A-8B-5I=\begin{bmatrix}15 & 20\\ 6 & 7\end{bmatrix}$.

$\det(A^3-B^3)=105-120=-15$.

For 2×2 matrix: $\det(\operatorname{adj}(M))=\det(M)$.

$(\det(\operatorname{adj}(A^3-B^3)))^2=225$.

11. **(3)** From $B=(I+A)^{-1}$ we get $B(I+A)=I$, so $BA=I-B$.

From $A+C=I$: $C=I-A$.

Calculating $CB=C(I+A)^{-1}=(I-A)(I+A)^{-1}$.

Since $B(I+A)=I$, we have $(I+A)=B^{-1}$, so $C=(I-A)B^{-1}$.

Multiplying by B: $CB=(I-A)$.

However, using the given $BC=\begin{bmatrix}1 & -5\\ -1 & 2\end{bmatrix}$ and $B(I+A)=I$ implies

$B-BA=B-(I-B)=2B-I=BC$, i.e. $BC=2B-I$.

Therefore $B=\frac{1}{2}(BC+I)=\frac{1}{2}\begin{bmatrix}2 & -5\\ -1 & 3\end{bmatrix}=\begin{bmatrix}1 & -5/2\\ -1/2 & 3/2\end{bmatrix}$.

Also $CB=2B-I=\begin{bmatrix}1 & -5\\ -1 & 2\end{bmatrix}$.

From $CB\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}12\\ -6\end{bmatrix}$: $(CB)^{-1}=-\frac{1}{3}\begin{bmatrix}2 & 5\\ 1 & 1\end{bmatrix}$.

So $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\frac{1}{3}\begin{bmatrix} -6 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$.
Therefore $x_1 + x_2 = 0$

12. (312) For a 3×2 matrix A, the diagonal sum of $A^T A$ equals the sum of squares of all 6 elements of A. We need: $a_{11}^2 + a_{21}^2 + a_{31}^2 + a_{12}^2 + a_{22}^2 + a_{32}^2 = 5$ where each element is from $\{-2, -1, 0, 1, 2\}$.
Possible squares: $0^2 = 0$, $(\pm 1)^2 = 1$, $(\pm 2)^2 = 4$.
Case 1:
Distribution $4 + 1 + 0 + 0 + 0 + 0$ (one ±2, one ±1, four 0's).
Arrangements: $\binom{6}{1} \times \binom{5}{1} = 30$.
Sign choices: $2 \times 2 = 4$. Total: $30 \times 4 = 120$.
Case 2:
Distribution $1 + 1 + 1 + 1 + 1 + 0$ (five ±1, one 0).
Arrangements: $\binom{6}{1} = 6$ ways to place the 0.
Sign choices for five ±1: $2^5 = 32$. Total: $6 \times 32 = 192$.
Total matrices: $120 + 192 = 312$